

Quantitative Methods for Business and Economics

Lecture 1 - Introduction and foundational concepts

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Introduction: Mathematics for Business and Economics

- Mathematics per se **not** a distinct field in business and economics
- We use mathematics **as an approach** to study reality
- It is not hard to think about business and economic models which were initially not mathematical
 - Consumer choice through utility maximization
 - Firm pricing and profit maximization
 - Schumpeter's model of creative destruction
- Modern researchers represent models using math
 - Allows to extend the model using mathematical “laws”
 - Makes models less subject to interpretation
 - Allows for statistical testing of hypotheses

Is it so challenging?

- Math is often feared because of its complexity
- Often the reason people fail is poor foundations
- Must master basic concepts before moving to more complex ones
 - This means that math takes time and few have the patience for it
 - Comparing yourself to others is **not useful**
 - Some have better foundations, some have inherent talent, some have more time

Order of operations

- The order of operations defines the priority of operations in an expression
- This is a consensus which everyone agreed to so we can all understand each other
 - Calculators and statistical software follow this order
 - As basic as this is, this is tested on the GRE (graduate school entrance exam)
- **PEMDAS: P**arentheses, **E**xponents, **M**ultiplication and **D**ivision, **A**ddition and **S**ubtraction
- If two operations have the same priority, we go from left to right.
- No sign means multiplication

Exercise

Order of operations

Calculate the following expression:

$$3 + 4 \times 2^2(3 + 1 \times 3 \div 3) + 4 - 5 + 5$$

Solution

Order of operations

- Start with the parentheses. Apply PEMDAS within too.

$$4 \times 1 = 4$$

- Substitute into the full expression, follow order, and solve

$$3 + 4 \times 2^2(4) + 4 - 5 + 5$$

$$3 + 4 \times 4(4) + 4 - 5 + 5$$

$$3 + 4 \times 16 + 4 - 5 + 5$$

$$3 + 64 + 4 - 5 + 5$$

$$67 + 4 - 5 + 5$$

$$71 - 5 + 5$$

$$71$$

Thoughts

- As useless this might be with numbers, it is crucial when working with variables
 - E.g. What does $\lambda_1(x_1 + x_2) \times \lambda_1 x_1 + x_2^2(\tau \cdot \gamma)$ simplify to?
- Calculators typically don't work with variables
- Not understanding PEMDAS may lead you to code the wrong formula in your software
 - It is hard to debug this as it not a syntax error

Variables and the greek alphabet

- Variables are used to represent unknowns in equations
 - Technically, they are placeholders for numbers
- x is commonly used, y and z are common too
- Scientists need more variables as they study more complex systems
 - The greek alphabet is used for this
- Get comfortable seeing and using the greek alphabet

The Greek alphabet

Alpha - Α α	Eta - Η η	Nu - Ν ν	Tau - Τ τ
Beta - Β β	Theta - Θ θ	Xi - Ξ ξ	Upsilon - Υ υ
Gamma - Γ γ	Iota - Ι ι	Omicron - Ο ο	Phi - Φ φ
Delta - Δ δ	Kappa - Κ κ	Pi - Π π	Chi - Χ χ
Epsilon - Ε ε	Lambda - Λ λ	Rho - Ρ ρ	Psi - Ψ ψ
Zeta - Ζ ζ	Mu - Μ μ	Sigma - Σ σ/ς	Omega - Ω ω

Figure 1: Greek alphabet. Source: University of Northern Colorado

- Commonly, the same letter is used for different purposes in different fields and contexts
 - β represents the slope coefficient, but also the discount factor in finance

Working with percentages and proportions

- Percentages are a way to express a fraction of a whole (the symbol being %)
- The word “percent” means “per hundred”; meaningful for calculation

$$\text{Percentage}\% = \frac{\text{Part}}{\text{Whole}} \times 100\%$$

- A useful way of thinking about percentages is as *proportions*: the decimal representation of the percentage
 - Divide by 100 and drop the % sign
 - 50% is 0.50, 25% is 0.25, 100% is 1.00
- This is useful for calculations so you don't use the above formula all the time
- Example: A 25% discount on a \$75 item is $0.25 \times 75 = 18.75$

Working with percentages and proportions

- Example: What is the final price of a \$75 item with a 25% discount?
- You may be tempted to use the rule of three, but this is not necessary (it takes longer)
- Use proportions to understand that 1 represents the full price (the \$75) and 0.25 represents the discount.

$$\$75 \cdot (1 - 0.25) = \$75 \cdot 0.75 = \$56.25$$

- Notice that the new final price is 75% of the original price, so this makes sense!
 - Don't be afraid to use your intuition
 - Business analysts and economists are not math machines, we use intuition to guide us

Percentage change and subindices

- It is common to express increases and decreases in percentages
- This involves introducing the concept of **variation**
 - Typically, we express change or variation as Δ
- Often, we use subindices to represent the old and new values
 - E.g. x_0 is the old value, x_1 is the new value
- The formula for percentage change is

$$\Delta\% = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100\% = \frac{x_1 - x_0}{x_0} \times 100$$

- A shorthand for this is dividing the final value by the initial value and subtracting 1
 - A negative value means a decrease, a positive value means an increase

Solving for a final value after a percentage change

- Example: A company gives employees a 10% raise. If an employee currently earns \$46,000 per year, what would the new salary be?
- Typically, one would calculate 10% of \$46,000 and add it to the original value.

$$x_0 + (x_0 \cdot \Delta\%) = x_1$$

- It is faster to represent $\Delta\%$ as a proportion, add it to 1, and multiply by the original value ($1 + 0.10 = 1.10$)
- Doing this is a simplification from the formula above
- However, it is also intuitive, since you're simply finding a value which is 10% larger than the whole (which is 1).
- So, the new salary is $\$46,000 \times 1.10 = \$50,600$

Solving for a final value after a percentage change

- The same can be done for percentage *decreases*
- Example: A store's weekly revenue has decreased by 5%. If revenue was \$500, what is the new weekly revenue?
- The new weekly revenue is $\$500 \times (1-0.05) = \$500 \times 0.95 = \$475$
- Once again, this is intuitive: the new revenue is only 95% of the original revenue

An exercise for you

- Example: A coffee drink in Toronto costs CAD 3.50 after a 15% sales tax. How much tax is the consumer paying?

Solution

- The current tax is 15% of the pre-tax price
- In many Canadian retail settings, taxes are added based on the pre-tax price
- The price after tax can be calculated as $P_{bf} \cdot 1.15 = 3.50$, so it is $3.50/1.15 \approx 3.04$
- The tax is the difference between the pre-tax and final price, so it is $3.50 - 3.04 = 0.46$
- Notice that if I had calculated 15% of \$3.50, I would have gotten \$0.525, which is not the same
- This is because the tax is calculated on the price before tax, not the price after tax.
- You have to be very careful when defining the “whole” in the percentage formula!

Ratios

- Ratios are a way to express the relationship between two parts of a whole
 - Not the same as the percentage, which is the relationship of a part to the whole
- They are often expressed as 1 to 2, 1:2.
- These can be a bit confusing, but context is helpful.
 - A ratio of 1:2 means that for every 1 unit of the first part, there are 2 units of the second part
- Example: A production function using a ratio of 1:2 for labor and capital means that for every 1 unit of labor, 2 units of capital are used.

Ratios

- You can work out the the relationship from one part to the whole by adding the parts of the ratio.
 - This may or may not make sense depending on the context
- With a ratio of 1:2 for labor and capital, the whole is 3 units, but this is not meaningful
- However, if you're told that in a hospital the ratio of doctors to nurses is 1:2, then doctors are $\frac{1}{3}$ of the whole staff

Application: sales tax changes and percentage change interpretations

- Suppose a sales tax rate rises from 12% to 15%
- Question for you: which the following statements is correct?
 - 1 There was a 3 percentage point increase in the sales tax rate
 - 2 There was a 25% increase in the sales tax rate
 - 3 There was a 15% increase in the sales tax rate

Percentage points and percent changes

- When we are analyzing something is measured in percentages itself, we need to be careful
- We call the the absolute change for these cases a **percentage point change**
- This is different from a **percent change**, which is the relative change from a value
- It might or might not be useful to use percentage changes for these cases, as it can be misleading
- This is when expressing everything as a proportion might be useful.

Equations

- An equation is a statement that two expressions are equal, typically with an unknown.
 - E.g. $x + 2 = 5$
- We often care about solving equations, which means finding the value of the unknown that makes the equation true (i.e. satisfies the equation)
 - For above, $x = 3$ will make the equation true ($5 = 5$)
- Equations can be solved by isolating the unknown on one side of the equation, and doing the math.

Equations

- There are rules for solving equations:
 - You can add or subtract the same value to both sides
 - You can multiply or divide both sides by the same value
 - Exponents must be raised to the same power on both sides
 - You can take the square root of both sides by adding $\pm$
 - We will see more rules as we go along

Functions

- A function is a rule that assigns a unique output to each input
 - Think of it as a machine that takes an input and gives an output
 - We stop caring about solving for the unknown, and care about the relationship between the input and output
- Functions are often represented as $f(x)$, where f is the function and x is the input
 - The value of the function at x is $f(x)$
- A function doesn't necessarily have to be a mathematical formula, but it is often represented as one
- Economists and business analysts use functions to represent real-world relationships
 - E.g. the relationship between advertising and sales
 - E.g. the relationship between price and quantity demanded
- Functions can be linear, quadratic, exponential, logarithmic, etc.

Functions

- The simplest function is the constant function, which always returns the same value
 - E.g. $f(x) = 5$
- Notice how this challenges our equation solving skills
 - There is no unknown to solve for
 - If we input 3, we get 5, if we input 10, we get 5
- A linear function is the next simplest.
 - E.g. $f(x) = 2x + 3$
 - Here we do have an input x which goes into the function
 - The function multiplies the input by 2, adds 3, and returns the result
 - It's called linear because the graph of the function is a straight line

Functions: domain and range

- The **domain** of a function is the set of all possible inputs
 - For $f(x) = 2x + 3$, the domain is all real numbers ($\mathbb{R}$)
- The **range** of a function is the set of all possible outputs
 - For $f(x) = 2x + 3$, the range is all real numbers ($\mathbb{R}$)
- The domain and range are important to understand the behavior of the function

Functions: domain and range

- Some functions have a restricted domain because of the way the equation is defined
 - E.g. $f(x) = \frac{1}{x}$ has a domain of all real numbers except 0
 - The function is not defined at 0 because there is no such thing as division by 0
- The domain and range are often represented as intervals
 - E.g. $x \in (-\infty, 0) \cup (0, \infty)$
 - E.g. $f(x) \in (-\infty, 0) \cup (0, \infty)$
- Without differential calculus, to find the domain and range of a function easily, you can plot it
 - With some time, you will start recognizing the behavior of functions

Inequalities

- Inequalities are statements that two expressions are not equal
 - E.g. $x > 5$
- When we have more elaborate expressions, we can have more complex inequalities
 - E.g. $2x + 3 > 5$
 - Simplify using the rules of equations to find which values of x make the inequality true.
 - E.g. $2x > 2$, $x > 1$. This means that all values of x greater than 1 make $2x + 3 > 5$ true.
- Use the same rules as equations to solve inequalities, except when multiplying or dividing by a negative number
 - In this case, the inequality sign flips
 - E.g. $-2x < 6$, $x > -3$
- Inequalities are often used to represent constraints in optimization problems

Application: returns to scale in a production function

- A production function is a mathematical model of how a country, a firm or any economic agent transform resources (inputs) into goods and services (outputs)
- In certain contexts, researchers care about whether the production function can be scaled
 - This means asking whether doubling the inputs will double the outputs?
 - This is called **returns to scale**
- This can be worked out by combining all of our knowledge so far

Application: returns to scale in a production function

- A production function is given by $Q = 2L + 3K$, where Q is the output, L is labor, and K is capital.
 - Think of it as a firm that produces goods (can be pizza) using workers and ovens.
- What happens to output if we double both labor and capital? Can the pizza firm double pizza production?
- We can work this out by plugging in values to the production function.
 - We need a base value of output. Let us see what happens when $L = 1$ and $K = 1$.
 - $Q = 2(1) + 3(1) = 2 + 3 = 5$
- Now, let us see what happens when $L = 2$ and $K = 2$.
 - $Q = 2(2) + 3(2) = 4 + 6 = 10$

Application: returns to scale in a production function

- How to see if output has doubled? We can calculate the percentage change in output.
 - $\Delta\% = 10/5 - 1 = 1$
 - This means that output changed by 100% (doubled)
- When output changes by the same proportion as the inputs, we say that the production function has constant returns to scale.
 - If you double the inputs, you double the outputs

Application: returns to scale in a production function

- If the percentage change in output is less than the percentage change in inputs, we say that the production function has decreasing returns to scale.
 - If you double the inputs, you get less than double the outputs
- If the percentage change in output is more than the percentage change in inputs, we say that the production function has increasing returns to scale.
 - If you double the inputs, you get more than double the outputs
- Returns to scale are important for understanding the behavior of firms and industries.
 - What kind of production function do you think a fast-food chain might have?
 - How do you think analysts can estimate the production function of a firm?